EXPERIMENT NO:09

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AIM: Implement a search functionality to filter API results dynamically.

Program:

```
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <title>API Search Demo</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600&di
splay=swap" rel="stylesheet">

  <style>
    :root {
      --bg: #0f172a;
      --bg-soft: #020617;
      --accent: #3b82f6;
      --accent-soft: rgba(59,130,246,0.15);
      --card-bg: #020617;
      --text-main: #e5e7eb;
      --text-muted: #9ca3af;
      --border-subtle: rgba(148,163,184,0.4);
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: "Poppins", system-ui, -apple-system, BlinkMacSystemFont,
"Segoe UI", sans-serif;
      min-height: 100vh;
      display: flex;
```

```
      align-items: center;
      justify-content: center;
      background: radial-gradient(circle at top, #1e293b, #020617);
      color: var(--text-main);
      padding: 24px;
    }

    .app {
      width: 100%;
      max-width: 1100px;
      background: linear-gradient(135deg, rgba(15,23,42,0.96),
rgba(15,23,42,0.98));
      border-radius: 18px;
      border: 1px solid rgba(148,163,184,0.35);
      box-shadow:
        0 18px 45px rgba(15,23,42,0.7),
        0 0 0 1px rgba(148,163,184,0.1);
      padding: 24px 26px 22px;
      backdrop-filter: blur(12px);
    }

    .header {
      display: flex;
      justify-content: space-between;
      align-items: flex-start;
      gap: 12px;
      margin-bottom: 18px;
    }

    .title-block h1 {
      font-size: 22px;
      font-weight: 600;
      letter-spacing: 0.02em;
      display: flex;
      align-items: center;
      gap: 8px;
    }

    .title-pill {
      font-size: 11px;
      padding: 2px 8px;
      border-radius: 999px;
      border: 1px solid rgba(52,211,153,0.6);
      color: #6ee7b7;
      background: rgba(6,95,70,0.25);
    }

    .title-block p {
```

```
  margin-top: 4px;
  font-size: 13px;
  color: var(--text-muted);
}

.badge {
  font-size: 11px;
  padding: 4px 10px;
  border-radius: 999px;
  border: 1px solid rgba(59,130,246,0.45);
  color: #bfdbfe;
  background: rgba(15,23,42,0.9);
  display: flex;
  align-items: center;
  gap: 6px;
  white-space: nowrap;
}

.badge-dot {
  width: 6px;
  height: 6px;
  border-radius: 999px;
  background: #4ade80;
  box-shadow: 0 0 0 4px rgba(74,222,128,0.25);
}

.search-wrapper {
  margin-bottom: 16px;
}

.search-label-row {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 6px;
  font-size: 12px;
  color: var(--text-muted);
}

.search-label-row span {
  opacity: 0.95;
}

.hint {
  font-size: 11px;
  color: var(--text-muted);
  opacity: 0.8;
}
```

```
    .search-box {
      position: relative;
    }

    .search-box input {
      width: 100%;
      padding: 10px 38px;
      font-size: 14px;
      border-radius: 999px;
      border: 1px solid rgba(148,163,184,0.6);
      background:
        radial-gradient(circle at 0 0, rgba(59,130,246,0.18), transparent
55%),
        #020617;
      color: var(--text-main);
      outline: none;
      transition: border-color 0.18s ease, box-shadow 0.18s ease, background
0.2s ease, transform 0.08s ease;
      box-shadow: 0 8px 22px rgba(15,23,42,0.85);
    }

    .search-box input::placeholder {
      color: rgba(148,163,184,0.8);
    }

    .search-box input:focus {
      border-color: rgba(59,130,246,0.9);
      box-shadow:
        0 0 0 1px rgba(59,130,246,0.6),
        0 18px 40px rgba(15,23,42,0.95);
      background:
        radial-gradient(circle at 1% 1%, rgba(59,130,246,0.3), transparent
60%),
        #020617;
      transform: translateY(-1px);
    }

    .search-icon {
      position: absolute;
      left: 12px;
      top: 50%;
      transform: translateY(-50%);
      font-size: 15px;
      color: rgba(148,163,184,0.9);
    }

    .search-kbd {
```

```
      position: absolute;
      right: 12px;
      top: 50%;
      transform: translateY(-50%);
      font-size: 11px;
      color: rgba(148,163,184,0.9);
      border-radius: 6px;
      border: 1px solid rgba(148,163,184,0.5);
      padding: 1px 4px;
      background: rgba(15,23,42,0.9);
    }

    .meta-row {
      display: flex;
      justify-content: space-between;
      align-items: center;
      margin-bottom: 10px;
      margin-top: 4px;
      font-size: 11px;
      color: var(--text-muted);
    }

    .count-pill {
      padding: 2px 8px;
      border-radius: 999px;
      background: rgba(30,64,175,0.35);
      border: 1px solid rgba(59,130,246,0.5);
      color: #bfdbfe;
    }

    .results-wrapper {
      background: radial-gradient(circle at top left, rgba(15,23,42,0.35),
rgba(15,23,42,0.9));
      border-radius: 14px;
      border: 1px solid rgba(51,65,85,0.9);
      padding: 10px 10px 12px;
      max-height: 520px;
      overflow: hidden;
      display: flex;
      flex-direction: column;
    }

    .results-scroll {
      overflow-y: auto;
      padding-right: 4px;
      margin-right: -2px;
    }
```

```
    .results-scroll::-webkit-scrollbar {
      width: 6px;
    }

    .results-scroll::-webkit-scrollbar-track {
      background: transparent;
    }

    .results-scroll::-webkit-scrollbar-thumb {
      background: rgba(148,163,184,0.55);
      border-radius: 999px;
    }

    #results {
      display: grid;
      grid-template-columns: repeat(auto-fill, minmax(260px, 1fr));
      gap: 8px;
    }

    .card {
      position: relative;
      background: radial-gradient(circle at top right, rgba(30,64,175,0.18),
rgba(15,23,42,0.95));
      border-radius: 12px;
      border: 1px solid rgba(51,65,85,0.95);
      padding: 10px 11px 9px;
      box-shadow:
        0 10px 28px rgba(15,23,42,0.9),
        0 0 0 1px rgba(15,23,42,0.7);
      cursor: default;
      transition: transform 0.14s ease, box-shadow 0.14s ease, border-color
0.14s ease, background 0.16s ease;
    }

    .card:hover {
      transform: translateY(-2px) translateZ(0);
      box-shadow:
        0 20px 40px rgba(15,23,42,0.9),
        0 0 0 1px rgba(59,130,246,0.6);
      border-color: rgba(59,130,246,0.75);
      background: radial-gradient(circle at top right, rgba(59,130,246,0.22),
rgba(15,23,42,0.97));
    }

    .card-header-row {
      display: flex;
      justify-content: space-between;
      align-items: center;
```

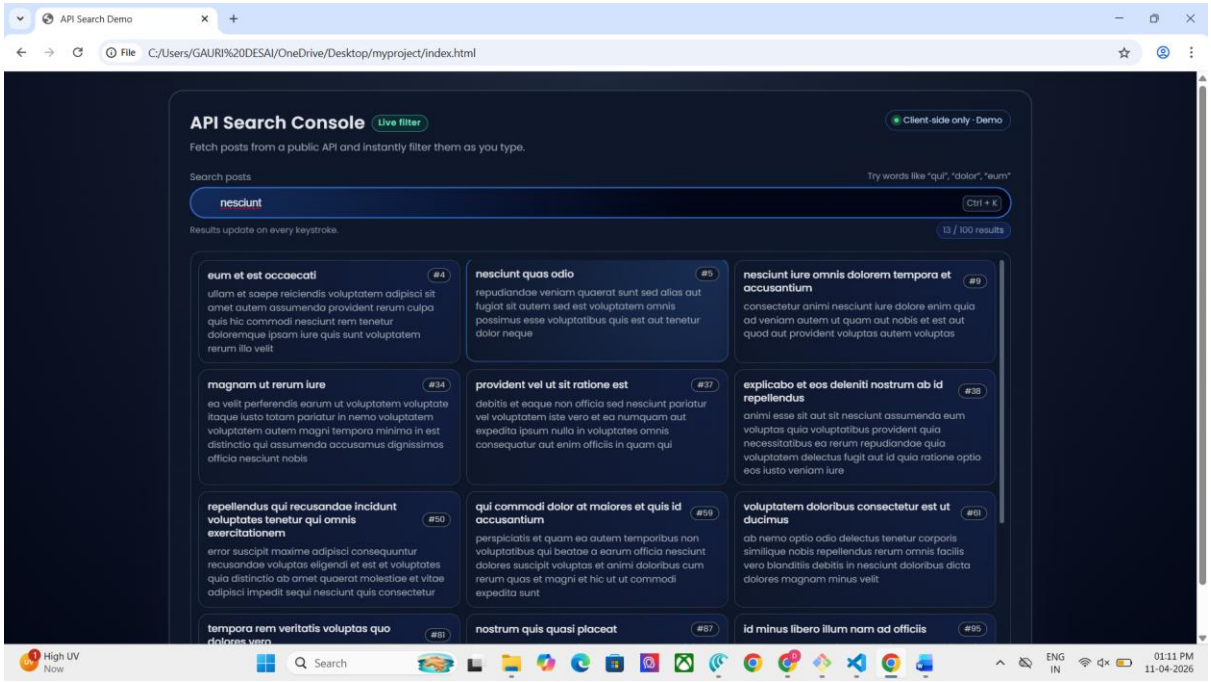
```
  gap: 4px;
  margin-bottom: 6px;
}

.card h3 {
  font-size: 13px;
  font-weight: 500;
  line-height: 1.3;
  color: #e5e7eb;
}

.chip-id {
  font-size: 10px;
  padding: 1px 7px;
  border-radius: 999px;
  border: 1px solid rgba(148,163,184,0.6);
  color: rgba(209,213,219,0.9);
  background: rgba(15,23,42,0.95);
}

.card p {
  font-size: 12px;
  color: var(--text-muted);
  line-height: 1.45;
}

.empty {
  font-size: 12px;
  color: var(--text-muted);
  padding: 12px 4px 4px;
  text-align: center;
}

@media (max-width: 720px) {
  .app {
    padding: 18px 16px 16px;
  }

  .header {
    flex-direction: column;
    align-items: flex-start;
    gap: 4px;
  }

  .results-wrapper {
    max-height: 70vh;
  }
}
```

```html
  </style>
</head>
<body>
  <main class="app">
    <header class="header">
      <div class="title-block">
        <h1>
          API Search Console
          <span class="title-pill">Live filter</span>
        </h1>
        <p>Fetch posts from a public API and instantly filter them as you
type.</p>
      </div>
      <div class="badge">
        <span class="badge-dot"></span>
        Client-side only · Demo
      </div>
    </header>

    <section class="search-wrapper">
      <div class="search-label-row">
        <span>Search posts</span>
        <span class="hint">Try words like “qui”, “dolor”, “eum”</span>
      </div>
      <div class="search-box">
        <span class="search-icon">🔍</span>
        <input
          type="text"
          id="search"
          placeholder="Type to filter by title or body..."
        />
        <span class="search-kbd">Ctrl + K</span>
      </div>
      <div class="meta-row">
        <span class="hint">Results update on every keystroke.</span>
        <span id="count" class="count-pill">Loading…</span>
      </div>
    </section>

    <section class="results-wrapper">
      <div class="results-scroll">
        <div id="results"></div>
        <p id="empty-message" class="empty" style="display:none;">
          No results match your search. Try a different keyword.
        </p>
      </div>
    </section>
  </main>
```

```
<script>
  const searchInput = document.getElementById('search');
  const resultsContainer = document.getElementById('results');
  const emptyMessage = document.getElementById('empty-message');
  const countEl = document.getElementById('count');

  let items = [];
  let debounceTimer;

  async function loadData() {
    try {
      const res = await fetch('https://jsonplaceholder.typicode.com/posts');
      items = await res.json();
      render(items);
      updateCount(items.length, items.length);
    } catch (err) {
      console.error('Error fetching data:', err);
      emptyMessage.textContent = 'Failed to load data from API.';
      emptyMessage.style.display = 'block';
      countEl.textContent = 'Error';
    }
  }

  function updateCount(visible, total) {
    countEl.textContent = `${visible} / ${total} results`;
  }

  function render(list) {
    if (!list.length) {
      resultsContainer.innerHTML = '';
      emptyMessage.style.display = 'block';
      return;
    }

    emptyMessage.style.display = 'none';

    resultsContainer.innerHTML = list.map(item => `
      <article class="card">
        <div class="card-header-row">
          <h3>${escapeHtml(item.title)}</h3>
          <span class="chip-id">#${item.id}</span>
        </div>
        <p>${escapeHtml(item.body)}</p>
      </article>
    `).join('');
  }
```

```
    function escapeHtml(str) {
      return String(str)
        .replace(/&/g, '&amp;')
        .replace(/</g, '&lt;')
        .replace(/>/g, '&gt;')
        .replace(/"/g, '&quot;')
        .replace(/'/g, '&#039;');
    }

    function handleSearch(value) {
      const term = value.toLowerCase().trim();

      const filtered = items.filter(item =>
        item.title.toLowerCase().includes(term) ||
        item.body.toLowerCase().includes(term)
      );

      render(filtered);
      updateCount(filtered.length, items.length);
    }

    searchInput.addEventListener('input', e => {
      clearTimeout(debounceTimer);
      const value = e.target.value;
      debounceTimer = setTimeout(() => {
        handleSearch(value);
      }, 180);
    });

    document.addEventListener('keydown', e => {
      if ((e.ctrlKey || e.metaKey) && e.key.toLowerCase() === 'k') {
        e.preventDefault();
        searchInput.focus();
        searchInput.select();
      }
    });

    loadData();
  </script>
</body>
</html>
```

OUTPUT:


API Search Demo
File C:/Users/GAURI%20DESAI/OneDrive/Desktop/myproject/index.html
API Search Console
Live filter
Client-side only · Demo
Fetch posts from a public API and instantly filter them as you type.
Search posts
Try words like "qui", "dolor", "eum"
nesciunt
Ctrl + K
Results update on every keystroke.
13 / 100 results
eum et est occaecati
#4
ullam et saepe reiciendis voluptatem adipisci sit amet autem assumenda provident rerum culpa quis hic commodi nesciunt rem tenetur doloremque ipsam iure quis sunt voluptatem rerum illo velit
nesciunt quas odio
#5
repudiandae veniam quaerat sunt sed alias aut fugiat sit autem sed est voluptatem omnis possimus esse voluptatibus quis est aut tenetur dolor neque
nesciunt iure omnis dolorem tempora et accusantium
#9
consectetur animi nesciunt iure dolore enim quia ad veniam autem ut quam aut nobis et est aut quod aut provident voluptas autem voluptas
magnam ut rerum iure
#34
ea velit perferendis earum ut voluptatem voluptate itaque iusto totam pariatur in nemo voluptatem voluptatem autem magni tempora minima in est distinctio qui assumenda accusamus dignissimos officia nesciunt nobis
provident vel ut sit ratione est
#37
debitis et eaque non officia sed nesciunt pariatur vel voluptatem iste vero et ea numquam aut expedita ipsum nulla in voluptates omnis consequatur aut enim officiis in quam qui
explicabo et eos deleniti nostrum ab id repellendus
#38
animi esse sit aut sit nesciunt assumenda eum voluptas quia voluptatibus provident quia necessitatibus ea rerum repudiandae quia voluptatem delectus fugit aut id quia ratione optio eos iusto veniam iure
repellendus qui recusandae incidunt voluptates tenetur qui omnis exercitationem
#50
error suscipit maxime adipisci consequuntur recusandae voluptas eligendi et est et voluptates quia distinctio ab amet quaerat molestiae et vitae adipisci impedit sequi nesciunt quis consectetur
qui commodi dolor at maiores et quis id accusantium
#59
perspiciatis et quam ea autem temporibus non voluptatibus qui beatae a earum officia nesciunt dolores suscipit voluptas et animi doloribus cum rerum quas et magni et hic ut ut commodi expedita sunt
voluptatem doloribus consectetur est ut ducimus
#61
ab nemo optio odio delectus tenetur corporis similique nobis repellendus rerum omnis facilis vero blanditiis debitis in nesciunt doloribus dicta dolores magnam minus velit
tempora rem veritatis voluptas quo
#81
nostrum quis quasi placeat
#87
id minus libero illum nam ad officiis
#95
High UV
Now
Search
ENG
IN
01:11 PM
11-04-2026